

Practica 7:

Firma con RSA escolar y CRT

Guevara Badillo Areli Alejandra
7CM1

Martes, 21 de abril, 2026

Código:

```
import math
import base64
from Crypto.Util.number import getPrime, inverse

def genKeys():
    p = getPrime(512)
    q = getPrime(512)

    n = p * q
    pi = (p - 1) * (q - 1)

    while True:
        e = getPrime(512)
        if math.gcd(e, pi) == 1:
            break

    d = inverse(e, pi)

    n_b64 = base64.b64encode(n.to_bytes((n.bit_length() + 7) // 8,
byteorder='big')).decode('utf-8')
    e_b64 = base64.b64encode(e.to_bytes((e.bit_length() + 7) // 8,
byteorder='big')).decode('utf-8')
    d_b64 = base64.b64encode(d.to_bytes((d.bit_length() + 7) // 8,
byteorder='big')).decode('utf-8')
    p_b64 = base64.b64encode(p.to_bytes((p.bit_length() + 7) // 8,
byteorder='big')).decode('utf-8')
    q_b64 = base64.b64encode(q.to_bytes((q.bit_length() + 7) // 8,
byteorder='big')).decode('utf-8')
    with open("Kpub.txt", "w") as f:
        f.write(f"{n_b64}\n{e_b64}")
    with open("Kpriv.txt", "w") as f:
        f.write(f"{n_b64}\n{d_b64}\n{p_b64}\n{q_b64}")

    return (e, n), (d, p, q)

def guardar(nombre, contenido):
```

```

    return

def main():
    (e, n), (d, p, q) = genKeys()

    print(f"e: {e}")
    print(f"n: {n}")

    return

if __name__ == "__main__":
    main()

```

Ejecución:

```

1 j1FBI6tn1UKuvR3IRue0wrE9XuFg7biuzpVQtqBIMmD9gEWuNGOpifG/MLsvxz1U8v2LL
2 tXronF7fb/A8u2Cc3SKrb2t4jb+ByoMXt40gSnw1R1lw1n84viwmHFBF3MhrJhcoAikmh
3 fZXzHb/ou7MYFq0ss/0cw144CqBh70zHzdRTXLf0XcpcphUgqD25PA6GH5EE2d5ZQ==
4 3c4fd56+9bQLEV02D4Zrz5x7cokwC+rakE+202JAmXZtCL3oZaZuULQwpHbgP70hQ==

```